

Research Assistant API

API Reference Documentation

v0.1.0

FastAPI service for scholarly discovery, metadata analytics,
semantic similarity, and user research workflows
over the Leeds OpenAlex subset.

Base URL <https://research-assistant-api.up.railway.app>
OpenAPI spec </openapi.json> | Interactive docs </docs>

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Overview

Introduction

The Research Assistant API provides programmatic access to a curated subset of the Leeds OpenAlex scholarly corpus. Endpoints cover paper search and discovery, author and topic lookup, citation graph traversal, semantic similarity, corpus analytics, and user-owned research workflow management including projects, reading lists, and annotations.

Base URL

```
https://research-assistant-api.up.railway.app
```

All paths below are relative to this base URL.

Authentication

Protected endpoints use **OAuth2 Password Bearer** tokens. Obtain a token from `POST /auth/register` or `POST /auth/login`, then pass it in every authenticated request:

```
Authorization: Bearer <your_access_token>
```

Endpoints requiring authentication are marked **Auth Required**. Requests to these endpoints without a valid token return **HTTP 401**.

Content Type

All request bodies and response bodies use `application/json`.

Pagination

List endpoints accept `limit` and `offset` query parameters.

Parameter	Type	Required	Description
<code>limit</code>	integer	No	Max results to return. Range 1–100. Default varies by endpoint.
<code>offset</code>	integer	No	Number of results to skip. Default: 0.

HTTP Status Codes

Code	Meaning	When returned
200	OK	Successful GET / PATCH request.
201	Created	Resource successfully created (POST).
204	No Content	Successful DELETE. No response body.
401	Unauthorized	Bearer token missing, invalid, or expired. Returns ErrorResponse .
404	Not Found	Resource does not exist or is not owned by the caller. Returns ErrorResponse .
409	Conflict	Request conflicts with current resource state (e.g. duplicate, missing embeddings). Returns ErrorResponse .
422	Unprocessable Entity	Request body or query parameter validation failed. Returns ValidationErrorResponse .

Error Response Shapes

All non-2xx responses return one of two shapes:

ErrorResponse — used for 401, 404, 409:

```
{ "detail": "Human-readable error message." }
```

ValidationErrorResponse — used for 422:

```
{
  "detail": [
    { "loc": ["body", "email"], "msg": "field required", "type": "missing" }
  ]
}
```

Health

Service health and liveness endpoints.

Health Check

GET /health Return a lightweight liveness response.

Responses

200 Returns a plain string-to-string object.

Example Request

```
curl https://research-assistant-api.up.railway.app/health
```

Example Response — 200 OK

```
{"status": "ok"}
```

Authentication

User registration, login, and bearer-token identity.

Register User — POST /auth/register

POST /auth/register Create a new user account and return a bearer access token.

Request Body — UserRegistrationRequest (required)

Parameter	Type	Required	Description
email	string	Yes	User email address. Length 3-255 characters.
password	string	Yes	Account password. Length 8-255 characters.

Responses

201 Account created. Returns AccessTokenResponse.

409 Email address already registered. Returns ErrorResponse.

422 Validation failed. Returns ValidationErrorResponse.

Example Request

```
curl -X POST https://research-assistant-api.up.railway.app/auth/register \
-H "Content-Type: application/json" \
-d '{"email": "researcher@leeds.ac.uk", "password": "securepass123"}'
```

Example Response — 201 Created

```
{
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJ1c2VyLTEyMyJ9.abc",
  "token_type": "bearer",
  "user": {
    "id": "550e8400-e29b-41d4-a716-446655440000",
    "email": "researcher@leeds.ac.uk",
    "created_at": "2024-06-01T09:00:00Z"
  }
}
```

Example Response — 409 Conflict

```
{"detail": "A user with this email already exists."}
```

Log In User — POST /auth/login

POST /auth/login Authenticate an existing user and return a bearer access token.

Request Body — UserLoginRequest (required)

Parameter	Type	Required	Description
email	string	Yes	Registered email address. Length 3-255 characters.
password	string	Yes	Account password. Length 8-255 characters.

Responses

200 Login successful. Returns AccessTokenResponse.

401 Invalid email or password. Returns ErrorResponse.

422 Validation failed. Returns ValidationErrorResponse.

Example Request

```
curl -X POST https://research-assistant-api.up.railway.app/auth/login \
-H "Content-Type: application/json" \
-d '{"email": "researcher@leeds.ac.uk", "password": "securepass123"}'
```

Example Response — 200 OK

```
{
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJ1c2VyLTEyMyJ9.abc",
  "token_type": "bearer",
  "user": {
    "id": "550e8400-e29b-41d4-a716-446655440000",
    "email": "researcher@leeds.ac.uk",
    "created_at": "2024-06-01T09:00:00Z"
  }
}
```

Example Response — 401 Unauthorized

```
{"detail": "Invalid email or password."}
```

Get Current User — GET /auth/me

GET /auth/me **Auth Required** Return the authenticated user linked to the bearer token.

Responses

200 Returns AuthenticatedUser.

401 Bearer token missing or invalid. Returns ErrorResponse.

Example Request

```
curl https://research-assistant-api.up.railway.app/auth/me \
-H "Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."
```

Example Response — 200 OK

```
{
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "email": "researcher@leeds.ac.uk",
  "created_at": "2024-06-01T09:00:00Z"
}
```

Analytics

Corpus-level analytics, top-paper ranking, and publication trends.

List Top Papers — GET /analytics/top-papers

GET /analytics/top-papers Rank papers by citation count with optional topic and year filters.

Query Parameters

Parameter	Type	Required	Description
topic	string	No	Filter by topic name.
year	integer	No	Filter by publication year (≥ 0).
limit	integer	No	Range 1–100. Default: 20.
offset	integer	No	Default: 0.

Responses

200 Returns array of AnalyticsPaperItem, ordered by descending citation count.

422 Validation failed. Returns ValidationErrorResponse.

Example Request

```
curl "https://research-assistant-api.up.railway.app/analytics/top-papers?topic=machine+learning&limit=2"
```

Example Response — 200 OK

```
[
  {
    "id": "W2741809807",
    "title": "Attention Is All You Need",
    "publication_year": 2017,
    "citation_count": 94821,
    "journal": "Advances in Neural Information Processing Systems",
    "topic": {"id": "T11413", "name": "Machine Learning", "field": "Computer Science"}
  },
  {
    "id": "W2963403868",
    "title": "BERT: Pre-training of Deep Bidirectional Transformers",
    "publication_year": 2019,
    "citation_count": 67432,
    "journal": "NAACL",
    "topic": {"id": "T11413", "name": "Machine Learning", "field": "Computer Science"}
  }
]
```

Get Topic Distribution — GET /analytics/topics

GET /analytics/topics Summarise topic coverage and aggregate citation counts across the corpus.

Query Parameters

Parameter	Type	Required	Description
year	integer	No	Filter to a specific publication year (≥ 0).
limit	integer	No	Range 1–100. Default: 50.
offset	integer	No	Default: 0.

Responses

200 Returns array of TopicAnalyticsItem.

422 Validation failed. Returns ValidationErrorResponse.

Example Request

```
curl "https://research-assistant-api.up.railway.app/analytics/topics?limit=2"
```


Example Response — 200 OK

```
[
  {
    "id": "T11413",
    "name": "Machine Learning",
    "field": "Computer Science",
    "paper_count": 1482,
    "total_citation_count": 428900,
    "average_citation_count": 289.4
  },
  {
    "id": "T10924",
    "name": "Bioinformatics",
    "field": "Biology",
    "paper_count": 963,
    "total_citation_count": 211430,
    "average_citation_count": 219.6
  }
]
```

Get Publication Trends — GET /analytics/trends

GET /analytics/trends Summarise publication output and citation totals by year.

Query Parameters

Parameter	Type	Required	Description
start_year	integer	No	Earliest year to include (≥ 0).
end_year	integer	No	Latest year to include (≥ 0).

Responses

200 Returns array of PublicationTrendItem, ordered by year.

422 Validation failed. Returns ValidationErrorResponse.

Example Request

```
curl "https://research-assistant-api.up.railway.app/analytics/trends?start_year=2020&end_year=2022"
```

Example Response — 200 OK

```
[
  {"publication_year": 2020, "paper_count": 741, "total_citation_count": 18900,
   "average_citation_count": 25.5},
  {"publication_year": 2021, "paper_count": 893, "total_citation_count": 19420,
   "average_citation_count": 21.7},
  {"publication_year": 2022, "paper_count": 1021, "total_citation_count": 14310,
   "average_citation_count": 14.0}
]
```

Papers

Paper discovery, detail, semantic similarity, citations, and paper-scoped annotations.

Search Papers — GET /papers/search

GET /papers/search Search the Leeds corpus by free text, topic, year, and citation threshold.

Query Parameters

Parameter	Type	Required	Description
<code>query</code>	string	No	Free-text search query.
<code>topic</code>	string	No	Filter by topic name.
<code>year</code>	integer	No	Filter by publication year (≥ 0).
<code>citation_count</code>	integer	No	Minimum citation count threshold (≥ 0).
<code>limit</code>	integer	No	Range 1–100. Default: 20.
<code>offset</code>	integer	No	Default: 0.

Responses

200 Returns array of PaperSummary.

422 Validation failed. Returns `ValidationErrorResponse`.

Example Request

```
curl "https://research-assistant-api.up.railway.app/papers/search?query=transformer&year=2020&limit=2"
```

Example Response — 200 OK

```
[
  {
    "id": "W3107570476",
    "title": "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale",
    "publication_year": 2020,
    "publication_date": "2020-10-22",
    "citation_count": 31200,
    "doi": "10.48550/arXiv.2010.11929",
    "journal": "ICLR 2021",
    "language": "en",
    "work_type": "article",
    "topic": {"id": "T11413", "name": "Machine Learning", "field": "Computer Science"}
  }
]
```

Get Paper — GET /papers/{paper_id}

GET /papers/{paper_id} Return full paper record with metadata, authorship, topic, and citation count.

Path Parameters

Parameter	Type	Required	Description
<code>paper_id</code>	string	Yes	Unique paper identifier.

Responses

200 Returns PaperDetail.

404 Paper not found. Returns `ErrorResponse`.

422 Validation failed. Returns `ValidationErrorResponse`.

Example Request

```
curl https://research-assistant-api.up.railway.app/papers/W2741809807
```

Example Response — 200 OK

```
{
  "id": "W2741809807",
  "title": "Attention Is All You Need",
  "publication_year": 2017,
  "publication_date": "2017-06-12",
  "citation_count": 94821,
  "doi": "10.48550/arXiv.1706.03762",
  "journal": "Advances in Neural Information Processing Systems",
  "language": "en",
  "work_type": "article",
  "topic": {"id": "T11413", "name": "Machine Learning", "field": "Computer Science"},
  "abstract": "The dominant sequence transduction models are based on complex recurrent...",
  "authors": [
    {
      "id": "A2109349281",
      "name": "Ashish Vaswani",
      "orcid": "0000-0002-5765-6641",
      "institution": {"id": "I1294671590", "name": "Google Brain", "country": "US"},
      "author_position": 1,
      "is_corresponding": true
    }
  ]
}
```

Example Response — 404 Not Found

```
{"detail": "Paper not found."}
```

List Similar Papers — GET /papers/{paper_id}/similar

GET /papers/{paper_id}/similar Return semantically similar papers using stored embeddings.

Path Parameters

Parameter	Type	Required	Description
<code>paper_id</code>	string	Yes	Source paper identifier.

Query Parameters

Parameter	Type	Required	Description
<code>limit</code>	integer	No	Range 1–100. Default: 10.
<code>offset</code>	integer	No	Default: 0.

Responses

- 200** Returns array of `SimilarPaperResponse`, ordered by descending similarity score.
- 404** Paper not found. Returns `ErrorResponse`.
- 409** No embeddings available for this paper. Returns `ErrorResponse`.
- 422** Validation failed. Returns `ValidationErrorResponse`.

Example Request

```
curl "https://research-assistant-api.up.railway.app/papers/W2741809807/similar?limit=2"
```

Example Response — 200 OK

```
[
  {
    "id": "W2963403868",
    "title": "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding",
    "publication_year": 2019,
    "publication_date": "2019-05-24",
    "citation_count": 67432,
    "doi": "10.18653/v1/N19-1423",
    "journal": "NAACL",
    "language": "en",
  }
]
```

```

    "work_type": "article",
    "topic": {"id": "T11413", "name": "Machine Learning", "field": "Computer Science"},
    "similarity_score": 0.9312
  }
]

```

Get Citation Neighbourhood — GET /papers/{paper_id}/citations

GET /papers/{paper_id}/citations Return papers cited by, and citing, the requested paper.

Path Parameters

Parameter	Type	Required	Description
<code>paper_id</code>	string	Yes	Central paper identifier.

Query Parameters

Parameter	Type	Required	Description
<code>limit</code>	integer	No	Range 1–100. Default: 20.
<code>offset</code>	integer	No	Default: 0.

Responses

- 200** Returns CitationNeighborhoodResponse.
- 404** Paper not found. Returns ErrorResponse.
- 422** Validation failed. Returns ValidationErrorResponse.

Example Request

```
curl "https://research-assistant-api.up.railway.app/papers/W2741809807/citations?limit=2"
```

Example Response — 200 OK

```

{
  "paper_id": "W2741809807",
  "cited_papers": [
    {
      "id": "W2128867445",
      "title": "Neural Machine Translation by Jointly Learning to Align and Translate",
      "publication_year": 2015,
      "citation_count": 21300,
      "doi": "10.48550/arXiv.1409.0473",
      "journal": null,
      "language": "en",
      "work_type": "article",
      "topic": {"id": "T11413", "name": "Machine Learning", "field": "Computer Science"}
    }
  ],
  "citing_papers": [
    {
      "id": "W2963403868",
      "title": "BERT: Pre-training of Deep Bidirectional Transformers",
      "publication_year": 2019,
      "citation_count": 67432,
      "doi": "10.18653/v1/N19-1423",
      "journal": "NAACL",
      "language": "en",
      "work_type": "article",
      "topic": {"id": "T11413", "name": "Machine Learning", "field": "Computer Science"}
    }
  ]
}

```

Get Citation Path — GET /papers/{paper_id}/path/{target_paper_id}

GET /papers/{paper_id}/path/{target_paper_id} Find the directed shortest citation path between two papers.

Path Parameters

Parameter	Type	Required	Description
<code>paper_id</code>	string	Yes	Source paper identifier.
<code>target_paper_id</code>	string	Yes	Target paper identifier.

Query Parameters

Parameter	Type	Required	Description
<code>max_depth</code>	integer	No	Maximum traversal depth. Range 1–12. Default: 6.

Responses

- 200** Returns CitationPathResponse.
- 404** One or both papers not found, or no path exists within `max_depth`. Returns ErrorResponse.
- 422** Validation failed. Returns ValidationErrorResponse.

Example Request

```
curl "https://research-assistant-api.up.railway.app/papers/W2741809807/path/W1981424483?max_depth=4"
```

Example Response — 200 OK

```
{
  "source_paper_id": "W2741809807",
  "target_paper_id": "W1981424483",
  "path_length": 2,
  "path": [
    {
      "id": "W2741809807",
      "title": "Attention Is All You Need",
      "publication_year": 2017,
      "citation_count": 94821,
      "doi": "10.48550/arXiv.1706.03762",
      "journal": "NeurIPS",
      "language": "en",
      "work_type": "article",
      "topic": {"id": "T11413", "name": "Machine Learning", "field": "Computer Science"}
    },
    {
      "id": "W1981424483",
      "title": "GPT-3: Language Models are Few-Shot Learners",
      "publication_year": 2020,
      "citation_count": 28000,
      "doi": "10.48550/arXiv.2005.14165",
      "journal": "NeurIPS",
      "language": "en",
      "work_type": "article",
      "topic": {"id": "T11413", "name": "Machine Learning", "field": "Computer Science"}
    }
  ]
}
```

Create Paper Annotation — POST /papers/{paper_id}/annotations

- POST** /papers/{paper_id}/annotations **Auth Required** Create a private annotation on a paper.

Path Parameters

Parameter	Type	Required	Description
<code>paper_id</code>	string	Yes	Paper to annotate.

Request Body — AnnotationCreateRequest (required)

Parameter	Type	Required	Description
text	string	Yes	Annotation content. Length 1–5000 characters.

Responses

- 201
- Annotation created. Returns
- `AnnotationResponse`
- .
- 401
- Bearer token missing or invalid. Returns
- `ErrorResponse`
- .
- 404
- Paper not found. Returns
- `ErrorResponse`
- .
- 422
- Validation failed. Returns
- `ValidationErrorResponse`
- .

Example Request

```
curl -X POST https://research-assistant-api.up.railway.app/papers/W2741809807/annotations \
-H "Authorization: Bearer eyJhbGci..." \
-H "Content-Type: application/json" \
-d '{"text": "Key paper for my literature review on attention mechanisms."}'
```

Example Response — 201 Created

```
{
  "id": "ann-7f3c8210-1a2b-4d5e-9f00-abc123def456",
  "user_id": "550e8400-e29b-41d4-a716-446655440000",
  "paper_id": "W2741809807",
  "text": "Key paper for my literature review on attention mechanisms.",
  "created_at": "2024-06-15T14:22:10Z"
}
```

List Paper Annotations — GET /papers/{paper_id}/annotations

- GET
- /papers/{paper_id}/annotations
- Auth Required
- List the authenticated user’s annotations for a paper.

Path Parameters

Parameter	Type	Required	Description
paper_id	string	Yes	Paper identifier.

Query Parameters

Parameter	Type	Required	Description
limit	integer	No	Range 1–100. Default: 20.
offset	integer	No	Default: 0.

Responses

- 200
- Returns array of
- `AnnotationResponse`
- .
- 401
- Bearer token missing or invalid. Returns
- `ErrorResponse`
- .
- 404
- Paper not found. Returns
- `ErrorResponse`
- .
- 422
- Validation failed. Returns
- `ValidationErrorResponse`
- .

Example Request

```
curl https://research-assistant-api.up.railway.app/papers/W2741809807/annotations \
-H "Authorization: Bearer eyJhbGci..."
```

Example Response — 200 OK

```
[
  {
    "id": "ann-7f3c8210-1a2b-4d5e-9f00-abc123def456",
    "user_id": "550e8400-e29b-41d4-a716-446655440000",
    "paper_id": "W2741809807",
    "text": "Key paper for my literature review on attention mechanisms.",
    "created_at": "2024-06-15T14:22:10Z"
  }
]
```

```
}  
]
```

Annotations

Private annotation read, update, and delete by annotation ID.

Get Annotation — GET /annotations/{annotation_id}

GET

/annotations/{annotation_id}

Auth Required

Fetch a single private annotation owned by the authenticated user.

Path Parameters

Parameter	Type	Required	Description
annotation_id	string	Yes	Annotation identifier.

Responses

- 200

Returns AnnotationResponse.
- 401

Bearer token missing or invalid. Returns ErrorResponse.
- 404

Annotation not found or not owned by the caller. Returns ErrorResponse.
- 422

Validation failed. Returns ValidationErrorResponse.

Example Request

```
curl https://research-assistant-api.up.railway.app/annotations/ann-7f3c8210-1a2b-4d5e-9f00-abc123def456 \
-H "Authorization: Bearer eyJhbGci..."
```

Example Response — 200 OK

```
{
  "id": "ann-7f3c8210-1a2b-4d5e-9f00-abc123def456",
  "user_id": "550e8400-e29b-41d4-a716-446655440000",
  "paper_id": "W2741809807",
  "text": "Key paper for my literature review on attention mechanisms.",
  "created_at": "2024-06-15T14:22:10Z"
}
```

Update Annotation — PATCH /annotations/{annotation_id}

PATCH

/annotations/{annotation_id}

Auth Required

Update the text of an annotation owned by the authenticated user.

Path Parameters

Parameter	Type	Required	Description
annotation_id	string	Yes	Annotation identifier.

Request Body — AnnotationUpdateRequest (required)

Parameter	Type	Required	Description
text	string	Yes	Updated annotation content. Length 1–5000 characters.

Responses

- 200

Returns updated AnnotationResponse.
- 401

Bearer token missing or invalid. Returns ErrorResponse.
- 404

Annotation not found or not owned by the caller. Returns ErrorResponse.
- 422

Validation failed. Returns ValidationErrorResponse.

Example Request


```
curl -X PATCH
  https://research-assistant-api.up.railway.app/annotations/ann-7f3c8210-1a2b-4d5e-9f00-abc123def456
\
-H "Authorization: Bearer eyJhbGci..." \
-H "Content-Type: application/json" \
-d '{"text": "Foundational paper      must cite in introduction and methodology."}'
```

Example Response — 200 OK

```
{
  "id": "ann-7f3c8210-1a2b-4d5e-9f00-abc123def456",
  "user_id": "550e8400-e29b-41d4-a716-446655440000",
  "paper_id": "W2741809807",
  "text": "Foundational paper -- must cite in introduction and methodology.",
  "created_at": "2024-06-15T14:22:10Z"
}
```

Delete Annotation — DELETE /annotations/{annotation_id}

DELETE

/annotations/{annotation_id}

Auth Required

Delete a private annotation owned by the authenticated user.

Path Parameters

Parameter	Type	Required	Description
annotation_id	string	Yes	Annotation identifier.

Responses

- 204** Annotation deleted. No response body.
- 401** Bearer token missing or invalid. Returns ErrorResponse.
- 404** Annotation not found or not owned by the caller. Returns ErrorResponse.
- 422** Validation failed. Returns ValidationErrorResponse.

Example Request

```
curl -X DELETE
  https://research-assistant-api.up.railway.app/annotations/ann-7f3c8210-1a2b-4d5e-9f00-abc123def456
\
-H "Authorization: Bearer eyJhbGci..."
```

Example Response — 204 No Content

```
(empty body)
```

Authors

Author search, detail, and paper lookup.

List Authors — GET /authors

GET /authors List authors in the Leeds corpus subset.

Query Parameters

Parameter	Type	Required	Description
limit	integer	No	Range 1–100. Default: 50.
offset	integer	No	Default: 0.

Responses

200 Returns array of AuthorDetail.

422 Validation failed. Returns ValidationErrorResponse.

Example Request

```
curl "https://research-assistant-api.up.railway.app/authors?limit=2"
```

Example Response — 200 OK

```
[
  {
    "id": "A2109349281",
    "name": "Ashish Vaswani",
    "orcid": "0000-0002-5765-6641",
    "institution": {"id": "I1294671590", "name": "Google Brain", "country": "US"},
    "paper_count": 14
  },
  {
    "id": "A2038004948",
    "name": "Geoffrey Hinton",
    "orcid": null,
    "institution": {"id": "I185261750", "name": "University of Toronto", "country": "CA"},
    "paper_count": 87
  }
]
```

Search Authors — GET /authors/search

GET /authors/search Search authors by name or author ID.

Query Parameters

Parameter	Type	Required	Description
query	string	Yes	Search term — name or author ID. Minimum length: 1.
limit	integer	No	Range 1–100. Default: 50.
offset	integer	No	Default: 0.

Responses

200 Returns array of AuthorDetail.

422 Validation failed. Returns ValidationErrorResponse.

Example Request

```
curl "https://research-assistant-api.up.railway.app/authors/search?query=Hinton&limit=2"
```

Example Response — 200 OK

```
[
  {
    "id": "A2038004948",
    "name": "Geoffrey Hinton",
```

```
{
  "orcid": null,
  "institution": {"id": "I185261750", "name": "University of Toronto", "country": "CA"},
  "paper_count": 87
}
```

Get Author — GET /authors/{author_id}

GET /authors/{author_id} Return author metadata and affiliation details.

Path Parameters

Parameter	Type	Required	Description
author_id	string	Yes	Author identifier.

Responses

- 200** Returns AuthorDetail.
- 404** Author not found. Returns ErrorResponse.
- 422** Validation failed. Returns ValidationErrorResponse.

Example Request

```
curl https://research-assistant-api.up.railway.app/authors/A2038004948
```

Example Response — 200 OK

```
{
  "id": "A2038004948",
  "name": "Geoffrey Hinton",
  "orcid": null,
  "institution": {"id": "I185261750", "name": "University of Toronto", "country": "CA"},
  "paper_count": 87
}
```

List Author Papers — GET /authors/{author_id}/papers

GET /authors/{author_id}/papers List papers associated with a specific author.

Path Parameters

Parameter	Type	Required	Description
author_id	string	Yes	Author identifier.

Query Parameters

Parameter	Type	Required	Description
limit	integer	No	Range 1–100. Default: 20.
offset	integer	No	Default: 0.

Responses

- 200** Returns array of PaperSummary.
- 404** Author not found. Returns ErrorResponse.
- 422** Validation failed. Returns ValidationErrorResponse.

Example Request

```
curl "https://research-assistant-api.up.railway.app/authors/A2038004948/papers?limit=1"
```

Example Response — 200 OK

```
[
  {
    "id": "W1585195840",
    "title": "ImageNet Classification with Deep Convolutional Neural Networks",
    "publication_year": 2012,
    "publication_date": "2012-12-03",
    "citation_count": 119650,
    "doi": "10.1145/3065386",
    "journal": "Communications of the ACM",
    "language": "en",
    "work_type": "article",
    "topic": {"id": "T11413", "name": "Machine Learning", "field": "Computer Science"}
  }
]
```

Topics

Topic lookup and topic-scoped paper listing.

List Topics — GET /topics

GET /topics List topics available in the Leeds corpus subset.

Query Parameters

Parameter	Type	Required	Description
limit	integer	No	Range 1–100. Default: 50.
offset	integer	No	Default: 0.

Responses

200 Returns array of TopicListItem.

422 Validation failed. Returns ValidationErrorResponse.

Example Request

```
curl "https://research-assistant-api.up.railway.app/topics?limit=2"
```

Example Response — 200 OK

```
[
  {
    "id": "T11413",
    "name": "Machine Learning",
    "field": "Computer Science",
    "paper_count": 1482
  },
  {
    "id": "T10924",
    "name": "Bioinformatics",
    "field": "Biology",
    "paper_count": 963
  }
]
```

List Topic Papers — GET /topics/{topic_id}/papers

GET /topics/{topic_id}/papers List papers assigned to a topic.

Path Parameters

Parameter	Type	Required	Description
topic_id	string	Yes	Topic identifier.

Query Parameters

Parameter	Type	Required	Description
limit	integer	No	Range 1–100. Default: 20.
offset	integer	No	Default: 0.

Responses

200 Returns array of PaperSummary.

404 Topic not found. Returns ErrorResponse.

422 Validation failed. Returns ValidationErrorResponse.

Example Request

```
curl "https://research-assistant-api.up.railway.app/topics/T11413/papers?limit=1"
```

Example Response — 200 OK

```
[
  {
    "id": "W2741809807",
    "title": "Attention Is All You Need",
    "publication_year": 2017,
    "publication_date": "2017-06-12",
    "citation_count": 94821
  }
]
```

```
"doi": "10.48550/arXiv.1706.03762",  
"journal": "NeurIPS",  
"language": "en",  
"work_type": "article",  
"topic": {"id": "T11413", "name": "Machine Learning", "field": "Computer Science"}  
}  
]
```

Projects

User-owned projects, reading lists, and recommendation workflows.

List Projects — GET /projects

GET

/projects

Auth Required

 List all projects owned by the authenticated user.

Responses

- 200

Returns array of ProjectResponse.
- 401

Bearer token missing or invalid. Returns ErrorResponse.

Example Request

```
curl https://research-assistant-api.up.railway.app/projects \
-H "Authorization: Bearer eyJhbGci..."
```

Example Response — 200 OK

```
[
  {
    "id": "proj-a1b2c3d4-e5f6-7890-abcd-ef1234567890",
    "user_id": "550e8400-e29b-41d4-a716-446655440000",
    "title": "Transformer Architecture Survey",
    "description": "Collecting papers for my MSc literature review.",
    "created_at": "2024-06-01T10:00:00Z"
  }
]
```

Create Project — POST /projects

POST

/projects

Auth Required

 Create a new user-owned research project.

Request Body — ProjectCreateRequest (required)

Parameter	Type	Required	Description
title	string	Yes	Project title. Length 1–255 characters.
description	string	No	Optional project description. Max 5000 characters.

Responses

- 201

Project created. Returns ProjectResponse.
- 401

Bearer token missing or invalid. Returns ErrorResponse.
- 422

Validation failed. Returns ValidationErrorResponse.

Example Request

```
curl -X POST https://research-assistant-api.up.railway.app/projects \
-H "Authorization: Bearer eyJhbGci..." \
-H "Content-Type: application/json" \
-d '{"title": "Transformer Architecture Survey", "description": "MSc literature review."}'
```

Example Response — 201 Created

```
{
  "id": "proj-a1b2c3d4-e5f6-7890-abcd-ef1234567890",
  "user_id": "550e8400-e29b-41d4-a716-446655440000",
  "title": "Transformer Architecture Survey",
  "description": "MSc literature review.",
  "created_at": "2024-06-01T10:00:00Z"
}
```

Get Project — GET /projects/{project_id}

GET

/projects/{project_id}

Auth Required

 Return a single project owned by the authenticated user.

Path Parameters

Parameter	Type	Required	Description
project_id	string	Yes	Project identifier.

Responses

- 200Returns ProjectResponse.
- 401Bearer token missing or invalid. Returns ErrorResponse.
- 404Project not found or not owned by the caller. Returns ErrorResponse.
- 422Validation failed. Returns ValidationErrorResponse.

Example Request

```
curl https://research-assistant-api.up.railway.app/projects/proj-a1b2c3d4-e5f6-7890-abcd-ef1234567890 \
-H "Authorization: Bearer eyJhbGci..."
```

Example Response — 200 OK

```
{
  "id": "proj-a1b2c3d4-e5f6-7890-abcd-ef1234567890",
  "user_id": "550e8400-e29b-41d4-a716-446655440000",
  "title": "Transformer Architecture Survey",
  "description": "MSc literature review.",
  "created_at": "2024-06-01T10:00:00Z"
}
```

Update Project — PATCH /projects/{project_id}

PATCH

/projects/{project_id}

Auth Required

Update a project’s title or description.

Path Parameters

Parameter	Type	Required	Description
project_id	string	Yes	Project identifier.

Request Body — ProjectUpdateRequest (required)

Parameter	Type	Required	Description
title	string	No	Updated title. Length 1–255 characters.
description	string	No	Updated description. Max 5000 characters.

Responses

- 200Returns updated ProjectResponse.
- 401Bearer token missing or invalid. Returns ErrorResponse.
- 404Project not found or not owned by the caller. Returns ErrorResponse.
- 422Validation failed. Returns ValidationErrorResponse.

Example Request

```
curl -X PATCH
https://research-assistant-api.up.railway.app/projects/proj-a1b2c3d4-e5f6-7890-abcd-ef1234567890 \
-H "Authorization: Bearer eyJhbGci..." \
-H "Content-Type: application/json" \
-d '{"title": "Transformer Survey      Final", "description": "Updated for submission."}'
```

Example Response — 200 OK

```
{
  "id": "proj-a1b2c3d4-e5f6-7890-abcd-ef1234567890",
  "user_id": "550e8400-e29b-41d4-a716-446655440000",
  "title": "Transformer Survey -- Final",
  "description": "Updated for submission.",
}
```



```
"created_at": "2024-06-01T10:00:00Z"
}
```

Delete Project — DELETE /projects/{project_id}

DELETE

/projects/{project_id}

Auth Required

Delete a project and all its dependent records.

Path Parameters

Parameter	Type	Required	Description
project_id	string	Yes	Project identifier.

Responses

- 204

Project deleted. No response body.
- 401

Bearer token missing or invalid. Returns ErrorResponse.
- 404

Project not found or not owned by the caller. Returns ErrorResponse.
- 422

Validation failed. Returns ValidationErrorResponse.

Example Request

```
curl -X DELETE
https://research-assistant-api.up.railway.app/projects/proj-a1b2c3d4-e5f6-7890-abcd-ef1234567890 \
-H "Authorization: Bearer eyJhbGci..."
```

Example Response — 204 No Content

```
(empty body)
```

Add Reading-List Item — POST /projects/{project_id}/reading-list

POST

/projects/{project_id}/reading-list

Auth Required

Add a paper to a project reading list.

Path Parameters

Parameter	Type	Required	Description
project_id	string	Yes	Project identifier.

Request Body — ReadingListItemCreateRequest (required)

Parameter	Type	Required	Description
paper_id	string	Yes	Paper to add. Length 1-255 characters.
priority	string	No	Priority: low, medium, or high. Default: medium.
notes	string	No	Optional notes. Max 5000 characters.

Responses

- 201

Item added. Returns ReadingListItemResponse.
- 401

Bearer token missing or invalid. Returns ErrorResponse.
- 404

Project or paper not found, or project not owned by caller. Returns ErrorResponse.
- 409

Paper already in this project's reading list. Returns ErrorResponse.
- 422

Validation failed. Returns ValidationErrorResponse.

Example Request

```
curl -X POST
https://research-assistant-api.up.railway.app/projects/proj-a1b2c3d4-e5f6-7890-abcd-ef1234567890/reading-list \
-H "Authorization: Bearer eyJhbGci..." \
-H "Content-Type: application/json" \
-d '{"paper_id": "W2741809807", "priority": "high", "notes": "Read before Thursday."}'
```

Example Response — 201 Created

```
{
  "id": "rli-99887766-aabb-ccdd-eeff-001122334455",
  "project_id": "proj-a1b2c3d4-e5f6-7890-abcd-ef1234567890",
  "paper_id": "W2741809807",
  "priority": "high",
  "notes": "Read before Thursday.",
  "created_at": "2024-06-10T08:45:00Z",
  "paper": {
    "id": "W2741809807",
    "title": "Attention Is All You Need",
    "publication_year": 2017,
    "citation_count": 94821,
    "doi": "10.48550/arXiv.1706.03762",
    "journal": "NeurIPS",
    "language": "en",
    "work_type": "article",
    "topic": {"id": "T11413", "name": "Machine Learning", "field": "Computer Science"}
  }
}
```

List Reading-List Items — GET /projects/{project_id}/reading-list

GET /projects/{project_id}/reading-list

Auth Required

List all reading-list items for a project.

Path Parameters

Parameter	Type	Required	Description
project_id	string	Yes	Project identifier.

Responses

- 200 Returns array of ReadingListItemResponse.
- 401 Bearer token missing or invalid. Returns ErrorResponse.
- 404 Project not found or not owned by caller. Returns ErrorResponse.
- 422 Validation failed. Returns ValidationErrorResponse.

Example Request

```
curl
  https://research-assistant-api.up.railway.app/projects/proj-a1b2c3d4-e5f6-7890-abcd-ef1234567890/reading-list
\
-H "Authorization: Bearer eyJhbGci..."
```

Example Response — 200 OK

```
[
  {
    "id": "rli-99887766-aabb-ccdd-eeff-001122334455",
    "project_id": "proj-a1b2c3d4-e5f6-7890-abcd-ef1234567890",
    "paper_id": "W2741809807",
    "priority": "high",
    "notes": "Read before Thursday.",
    "created_at": "2024-06-10T08:45:00Z",
    "paper": {
      "id": "W2741809807",
      "title": "Attention Is All You Need",
      "publication_year": 2017,
      "citation_count": 94821,
      "doi": "10.48550/arXiv.1706.03762",
      "journal": "NeurIPS",
      "language": "en",
      "work_type": "article",
      "topic": {"id": "T11413", "name": "Machine Learning", "field": "Computer Science"}
    }
  }
]
```

List Project Recommendations — GET /projects/{project_id}/recommendations

GET /projects/{project_id}/recommendations **Auth Required** Recommend papers based on the project's reading list using semantic, citation, or hybrid scoring.

Path Parameters

Parameter	Type	Required	Description
<code>project_id</code>	string	Yes	Project identifier.

Query Parameters

Parameter	Type	Required	Description
<code>mode</code>	string	No	Scoring mode: semantic, citation, or hybrid. Default: hybrid.
<code>semantic_weight</code>	number	No	Weight for semantic score. Range 0.0–1.0.
<code>citation_weight</code>	number	No	Weight for citation score. Range 0.0–1.0.
<code>limit</code>	integer	No	Range 1–100. Default: 10.
<code>offset</code>	integer	No	Default: 0.

Responses

- 200** Returns array of `ProjectRecommendationResponse`.
- 401** Bearer token missing or invalid. Returns `ErrorResponse`.
- 404** Project not found or not owned by caller. Returns `ErrorResponse`.
- 409** Reading list is empty — cannot generate recommendations. Returns `ErrorResponse`.
- 422** Validation failed. Returns `ValidationErrorResponse`.

Example Request

```
curl
  https://research-assistant-api.up.railway.app/projects/proj-a1b2c3d4-e5f6-7890-abcd-ef1234567890/recommendations
  -H "Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cGEiOiJ1bmRlcm9kaWciLCJ1b250aW50IjoiIn0="
```

Example Response — 200 OK

```
[
  {
    "id": "W2963403868",
    "title": "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding",
    "publication_year": 2019,
    "publication_date": "2019-05-24",
    "citation_count": 67432,
    "doi": "10.18653/v1/N19-1423",
    "journal": "NAACL",
    "language": "en",
    "work_type": "article",
    "topic": {"id": "T11413", "name": "Machine Learning", "field": "Computer Science"},
    "recommendation_score": 0.887,
    "semantic_score": 0.931,
    "citation_score": 0.712,
    "scoring_mode": "hybrid",
    "semantic_weight": 0.6,
    "citation_weight": 0.4
  }
]
```

Example Response — 409 Conflict

```
{
  "detail": "Reading list is empty. Add papers before requesting recommendations."
}
```

Reading List

Update and delete individual reading-list items by item ID.

Update Reading-List Item — PATCH /reading-list-items/{item_id}

PATCH

/reading-list-items/{item_id}

Auth Required

Update the priority or notes for a reading-list item.

Path Parameters

Parameter	Type	Required	Description
item_id	string	Yes	Reading-list item identifier.

Request Body — ReadingListItemUpdateRequest (required)

Parameter	Type	Required	Description
priority	string	No	Updated priority: low, medium, or high.
notes	string	No	Updated notes. Max 5000 characters.

Responses

- 200
- Returns updated ReadingListItemResponse.
- 401
- Bearer token missing or invalid. Returns ErrorResponse.
- 404
- Item not found or not owned by caller. Returns ErrorResponse.
- 422
- Validation failed. Returns ValidationErrorResponse.

Example Request

```
curl -X PATCH
  https://research-assistant-api.up.railway.app/reading-list-items/rli-99887766-aabb-ccdd-eeff-001122334455
  \
  -H "Authorization: Bearer eyJhbGci..." \
  -H "Content-Type: application/json" \
  -d '{"priority": "low", "notes": "Already read -- good background only."}'
```

Example Response — 200 OK

```
{
  "id": "rli-99887766-aabb-ccdd-eeff-001122334455",
  "project_id": "proj-a1b2c3d4-e5f6-7890-abcd-ef1234567890",
  "paper_id": "W2741809807",
  "priority": "low",
  "notes": "Already read -- good background only.",
  "created_at": "2024-06-10T08:45:00Z",
  "paper": {
    "id": "W2741809807",
    "title": "Attention Is All You Need",
    "publication_year": 2017,
    "citation_count": 94821,
    "doi": "10.48550/arXiv.1706.03762",
    "journal": "NeurIPS",
    "language": "en",
    "work_type": "article",
    "topic": {"id": "T11413", "name": "Machine Learning", "field": "Computer Science"}
  }
}
```

Delete Reading-List Item — DELETE /reading-list-items/{item_id}

DELETE

/reading-list-items/{item_id}

Auth Required

Delete a reading-list item owned by the authenticated user.

Path Parameters

Parameter	Type	Required	Description
item_id	string	Yes	Reading-list item identifier.

Responses

- 204** Item deleted. No response body.
- 401** Bearer token missing or invalid. Returns `ErrorResponse`.
- 404** Item not found or not owned by caller. Returns `ErrorResponse`.
- 422** Validation failed. Returns `ValidationErrorResponse`.

Example Request

```
curl -X DELETE
      https://research-assistant-api.up.railway.app/reading-list-items/rli-99887766-aabb-ccdd-eeff-001122334455
\
-H "Authorization: Bearer eyJhbGci..."
```

Example Response — 204 No Content

```
(empty body)
```

Schemas

All request body, response, and error schemas defined in the OpenAPI specification.

UserRegistrationRequest

Field	Type	Req.	Description
<code>email</code>	string	Yes	User email address. Length 3–255 characters.
<code>password</code>	string	Yes	Account password. Length 8–255 characters.

UserLoginRequest

Field	Type	Req.	Description
<code>email</code>	string	Yes	Registered email address. Length 3–255 characters.
<code>password</code>	string	Yes	Account password. Length 8–255 characters.

AccessTokenResponse

Field	Type	Req.	Description
<code>access_token</code>	string	Yes	Bearer token for authenticated requests.
<code>token_type</code>	string	No	Token scheme. Always "bearer".
<code>user</code>	AuthenticatedUser	Yes	The authenticated user record.

AuthenticatedUser

Field	Type	Req.	Description
<code>id</code>	string	Yes	User UUID.
<code>email</code>	string	Yes	User email address.
<code>created_at</code>	string (date-time)	Yes	Account creation timestamp (ISO 8601).

PaperSummary

Base paper record returned by list and search endpoints.

Field	Type	Req.	Description
<code>id</code>	string	Yes	Paper identifier.
<code>title</code>	string	Yes	Paper title.
<code>publication_year</code>	integer	Yes	Year of publication.
<code>publication_date</code>	string (date)	No	Full publication date (ISO 8601).
<code>citation_count</code>	integer	Yes	Total citation count.
<code>doi</code>	string	No	Digital Object Identifier.
<code>journal</code>	string	No	Journal or venue name.
<code>language</code>	string	No	Language code (e.g. "en").
<code>work_type</code>	string	No	Work type e.g. "article", "book-chapter".
<code>topic</code>	TopicSummary	No	Assigned topic.

PaperDetail

Full paper record — all *PaperSummary* fields plus:

Field	Type	Req.	Description
id	string	Yes	Paper identifier. <i>(from PaperSummary)</i>
title	string	Yes	Paper title. <i>(from PaperSummary)</i>
publication_year	integer	Yes	Year of publication. <i>(from PaperSummary)</i>
publication_date	string (date)	No	Full publication date. <i>(from PaperSummary)</i>
citation_count	integer	Yes	Total citation count. <i>(from PaperSummary)</i>
doi	string	No	Digital Object Identifier. <i>(from PaperSummary)</i>
journal	string	No	Journal or venue name. <i>(from PaperSummary)</i>
language	string	No	Language code. <i>(from PaperSummary)</i>
work_type	string	No	Work type. <i>(from PaperSummary)</i>
topic	TopicSummary	No	Assigned topic. <i>(from PaperSummary)</i>
abstract	string	No	Paper abstract.
authors	PaperAuthorSummary []	Yes	Ordered author list.

PaperAuthorSummary

Field	Type	Req.	Description
id	string	Yes	Author identifier.
name	string	Yes	Author display name.
orcid	string	No	ORCID identifier.
institution	InstitutionSummary	No	Affiliated institution.
author_position	integer	No	Position in author list (1-indexed).
is_corresponding	boolean	Yes	Whether this is the corresponding author.

SimilarPaperResponse

All *PaperSummary* fields plus:

Field	Type	Req.	Description
id	string	Yes	Paper identifier. <i>(from PaperSummary)</i>
title	string	Yes	Paper title. <i>(from PaperSummary)</i>
publication_year	integer	Yes	Year of publication. <i>(from PaperSummary)</i>
publication_date	string (date)	No	Full publication date. <i>(from PaperSummary)</i>
citation_count	integer	Yes	Total citation count. <i>(from PaperSummary)</i>
doi	string	No	DOI. <i>(from PaperSummary)</i>
journal	string	No	Journal name. <i>(from PaperSummary)</i>
language	string	No	Language code. <i>(from PaperSummary)</i>
work_type	string	No	Work type. <i>(from PaperSummary)</i>
topic	TopicSummary	No	Assigned topic. <i>(from PaperSummary)</i>
similarity_score	number	Yes	Cosine similarity in range [0, 1].

CitationNeighborhoodResponse

Field	Type	Req.	Description
<code>paper_id</code>	string	Yes	Central paper identifier.
<code>cited_papers</code>	PaperSummary[]	Yes	Papers cited by this paper.
<code>citing_papers</code>	PaperSummary[]	Yes	Papers that cite this paper.

CitationPathResponse

Field	Type	Req.	Description
<code>source_paper_id</code>	string	Yes	Source paper identifier.
<code>target_paper_id</code>	string	Yes	Target paper identifier.
<code>path_length</code>	integer	Yes	Number of hops in the path.
<code>path</code>	PaperSummary[]	Yes	Ordered papers from source to target.

AuthorDetail

Field	Type	Req.	Description
<code>id</code>	string	Yes	Author identifier.
<code>name</code>	string	Yes	Author display name.
<code>orcid</code>	string	No	ORCID identifier.
<code>institution</code>	InstitutionSummary	No	Affiliated institution.
<code>paper_count</code>	integer	Yes	Number of papers in the corpus.

InstitutionSummary

Field	Type	Req.	Description
<code>id</code>	string	Yes	Institution identifier.
<code>name</code>	string	Yes	Institution name.
<code>country</code>	string	No	ISO country code.

TopicSummary

Field	Type	Req.	Description
<code>id</code>	string	Yes	Topic identifier.
<code>name</code>	string	Yes	Topic name.
<code>field</code>	string	No	Broad research field.

TopicListItem

All *TopicSummary* fields plus:

Field	Type	Req.	Description
<code>id</code>	string	Yes	Topic identifier. <i>(from TopicSummary)</i>
<code>name</code>	string	Yes	Topic name. <i>(from TopicSummary)</i>
<code>field</code>	string	No	Research field. <i>(from TopicSummary)</i>
<code>paper_count</code>	integer	Yes	Number of papers assigned to this topic.

AnalyticsPaperItem

Field	Type	Req.	Description
<code>id</code>	string	Yes	Paper identifier.
<code>title</code>	string	Yes	Paper title.
<code>publication_year</code>	integer	Yes	Year of publication.
<code>citation_count</code>	integer	Yes	Total citation count.
<code>journal</code>	string	No	Journal or venue name.
<code>topic</code>	TopicSummary	No	Assigned topic.

TopicAnalyticsItem

Field	Type	Req.	Description
<code>id</code>	string	Yes	Topic identifier.
<code>name</code>	string	Yes	Topic name.
<code>field</code>	string	No	Broad research field.
<code>paper_count</code>	integer	Yes	Number of papers in this topic.
<code>total_citation_count</code>	integer	Yes	Sum of citations across all papers in this topic.
<code>average_citation_count</code>	number	Yes	Mean citation count per paper.

PublicationTrendItem

Field	Type	Req.	Description
<code>publication_year</code>	integer	Yes	Calendar year.
<code>paper_count</code>	integer	Yes	Papers published that year.
<code>total_citation_count</code>	integer	Yes	Sum of citations for papers from that year.
<code>average_citation_count</code>	number	Yes	Mean citation count per paper for that year.

AnnotationCreateRequest

Field	Type	Req.	Description
<code>text</code>	string	Yes	Annotation content. Length 1–5000 characters.

AnnotationUpdateRequest

Field	Type	Req.	Description
<code>text</code>	string	Yes	Updated annotation content. Length 1–5000 characters.

AnnotationResponse

Field	Type	Req.	Description
<code>id</code>	string	Yes	Annotation identifier.
<code>user_id</code>	string	Yes	Owning user identifier.
<code>paper_id</code>	string	Yes	Annotated paper identifier.
<code>text</code>	string	Yes	Annotation content.

Field	Type	Req.	Description
<code>created_at</code>	string (date-time)	Yes	Creation timestamp (ISO 8601).

ProjectCreateRequest

Field	Type	Req.	Description
<code>title</code>	string	Yes	Project title. Length 1–255 characters.
<code>description</code>	string	No	Optional description. Max 5000 characters.

ProjectUpdateRequest

Field	Type	Req.	Description
<code>title</code>	string	No	Updated title. Length 1–255 characters.
<code>description</code>	string	No	Updated description. Max 5000 characters.

ProjectResponse

Field	Type	Req.	Description
<code>id</code>	string	Yes	Project identifier.
<code>user_id</code>	string	Yes	Owning user identifier.
<code>title</code>	string	Yes	Project title.
<code>description</code>	string	No	Project description.
<code>created_at</code>	string (date-time)	Yes	Creation timestamp (ISO 8601).

ProjectRecommendationResponse

All *PaperSummary* fields plus scoring metadata.

Field	Type	Req.	Description
<code>id</code>	string	Yes	Paper identifier. <i>(from PaperSummary)</i>
<code>title</code>	string	Yes	Paper title. <i>(from PaperSummary)</i>
<code>publication_year</code>	integer	Yes	Year of publication. <i>(from PaperSummary)</i>
<code>publication_date</code>	string (date)	No	Full publication date. <i>(from PaperSummary)</i>
<code>citation_count</code>	integer	Yes	Total citation count. <i>(from PaperSummary)</i>
<code>doi</code>	string	No	DOI. <i>(from PaperSummary)</i>
<code>journal</code>	string	No	Journal name. <i>(from PaperSummary)</i>
<code>language</code>	string	No	Language code. <i>(from PaperSummary)</i>
<code>work_type</code>	string	No	Work type. <i>(from PaperSummary)</i>
<code>topic</code>	TopicSummary	No	Assigned topic. <i>(from PaperSummary)</i>
<code>recommendation_score</code>	number	Yes	Final composite score.
<code>semantic_score</code>	number	Yes	Semantic similarity component.
<code>citation_score</code>	number	Yes	Citation graph component.
<code>scoring_mode</code>	string (enum)	Yes	<code>semantic</code> , <code>citation</code> , or <code>hybrid</code> .
<code>semantic_weight</code>	number	Yes	Weight applied to semantic score.
<code>citation_weight</code>	number	Yes	Weight applied to citation score.

ReadingListItemCreateRequest

Field	Type	Req.	Description
<code>paper_id</code>	string	Yes	Paper to add. Length 1–255 characters.
<code>priority</code>	string (enum)	No	low, medium, or high. Default: medium.
<code>notes</code>	string	No	Optional notes. Max 5000 characters.

ReadingListItemUpdateRequest

Field	Type	Req.	Description
<code>priority</code>	string (enum)	No	Updated priority: low, medium, or high.
<code>notes</code>	string	No	Updated notes. Max 5000 characters.

ReadingListItemResponse

Field	Type	Req.	Description
<code>id</code>	string	Yes	Reading-list item identifier.
<code>project_id</code>	string	Yes	Parent project identifier.
<code>paper_id</code>	string	Yes	Paper identifier.
<code>priority</code>	string (enum)	Yes	low, medium, or high.
<code>notes</code>	string	No	User notes for this item.
<code>created_at</code>	string (date-time)	Yes	Creation timestamp (ISO 8601).
<code>paper</code>	PaperSummary	Yes	Embedded full paper summary.

ErrorResponse

Returned for HTTP 401, 404, and 409 responses.

Field	Type	Req.	Description
<code>detail</code>	string	Yes	Human-readable description of the error.

ValidationErrorResponse

Returned for HTTP 422 Unprocessable Entity responses.

Field	Type	Req.	Description
<code>detail</code>	ValidationErrorItem[]	Yes	Array of individual field-level validation failures.

ValidationErrorItem

A single field-level failure within *ValidationErrorResponse.detail*.

Field	Type	Req.	Description
<code>loc</code>	(string integer)[]	Yes	Location path to the failed field — e.g. ["body", "email"].
<code>msg</code>	string	Yes	Human-readable error message.
<code>type</code>	string	Yes	Machine-readable error type identifier.